

CWA SCIENCE CLASSES

Concept With Abhishek | Class 9 — Physics

Chapter 2: Force and Laws of Motion

1. Newton's First Law (Law of Inertia): An object remains in its state of rest or uniform motion unless acted upon by an unbalanced external force.
2. Inertia depends on mass — heavier objects have more inertia.
3. Newton's Second Law: $F = ma$. Rate of change of momentum is directly proportional to the applied force.
4. Momentum: $p = mv$. SI unit: kg m/s.
5. Newton's Third Law: For every action there is an equal and opposite reaction. Action-reaction pairs act on different bodies.
6. Conservation of Momentum: In absence of external force, total momentum before collision = total momentum after collision.
7. Practical examples: rocket propulsion, recoil of gun, walking, swimming — all explained using Newton's third law.